

Lab. 5 - Project Report

Try fitting uni-variate Gaussian models to the different features of the different classes of the project dataset. For each class, for each component of the feature vector of that class, compute the ML estimate for the parameters of a 1D Gaussian distribution. Plot the distribution density (remember that you have to exponentiate the log-density) on top of the normalized histogram (set density=True when creating the histogram, see Laboratory 2). What do you observe? Are there features for which the Gaussian densities provide a good fit? Are there features for which the Gaussian model seems significantly less accurate? Note: for this part of the project, since we are still performing some preliminary, qualitative analysis, you can compute the ML estimates and the plots either on the whole training set, or on the model training split of the training dataset. In the following labs we will employ the densities for classification, and we will need to perform model selection, therefore we will re-compute ML estimates on the model training portion of the dataset only (see Laboratory 3).

All data

Class1

Class 1 mu: [[0.00287744] [0.01869317] [-0.68093944] [0.6708359] [0.02795693] [-0.00582741]]

Class 1 C: [[5.6958103e-01 4.4340142e-03 8.4317056e-03 1.4447930e-02 1.4267275e-02 -1.4115002e-03] [4.4340142e-03 1.4208657e+00 -4.7580758e-03 -1.2243296e-02 -2.5728114e-03 -5.5409307e-03] [8.4317056e-03 -4.7580758e-03 5.4997694e-01 1.3203588e-02 -1.1525188e-02 1.1104402e-02] [1.4447930e-02 -1.2243296e-02 1.3203588e-02 5.3604269e-01 7.0738117e-03 1.5974429e-02] [1.4267275e-02 -2.5728114e-03 -1.1525188e-02 7.0738117e-03 6.8007350e-01 1.2000162e-02] [-1.4115002e-03 -5.5409307e-03 1.1104402e-02 1.5974429e-02 1.2000162e-02 7.0503855e-01]]

Difference between mu: 0.0

Difference between C: 0.0

Maximum Likelihood Estimation - ML: -22208.396989859666

Maximum Likelihood Estimation: -22208.396989859666

Log-Likelihood -22208.396989859666

Class 2 mu: [[5.4454483e-04] [-8.5243722e-03] [6.6523772e-01] [-6.6419560e-01] [-4.1725226e-02] [2.3938498e-02]]

Class2

Class 2 C: [[1.4302334 0.0058646 -0.00328748 0.01305324 0.00569338 -0.02439541] [0.0058646 0.57827795 -0.0085804 -0.006824 -0.00638891 0.00427788] [-0.00328748 -0.0085804 0.54890263 0.01126717 0.00465291 -0.01835332] [0.01305324 -0.006824 0.01126717 0.5533427 -0.00288255 0.02041833] [0.00569338 -0.00638891 0.00465291 -0.00288255 1.3177681 -0.00914105] [-0.02439541 0.00427788 -0.01835332 0.02041833 -0.00914105 1.2870262]]

Difference between mu: 0.0

Difference between C: 0.0

Maximum Likelihood Estimation - ML: -24337.94154930718

Maximum Likelihood Estimation: -24337.94154930718

Log-Likelihood -24337.94154930718

Training data Class 1

Class 1 mu: [[-0.00652448] [0.01606322] [-0.67409414] [0.6805161] [0.01566136] [-0.01328887]]

Class 1 C: [[6.0095656e-01 5.1567145e-05 1.9058922e-02 1.9252986e-02 1.2803944e-02 -1.3472160e-02] [5.1567145e-05 1.4472256e+00 -1.6134009e-02 -1.5856147e-02 -2.6452910e-02 2.2913983e-02] [1.9058922e-02 -1.6134009e-02 5.6534892e-01 -1.8434401e-03 -6.9144638e-03 1.6892832e-02] [1.9252986e-02 -1.5856147e-02 -1.8434401e-03 5.4161525e-01 5.2517131e-03 1.3571776e-02] [1.2803944e-02 -2.6452910e-02 -6.9144638e-03 5.2517131e-03 6.9606781e-01 1.5843840e-02] [-1.3472160e-02 2.2913983e-02 1.6892832e-02 1.3571776e-02 1.5843840e-02 6.8651968e-01]]

Difference between mu: 0.0

Difference between C: 0.0

Maximum Likelihood Estimation - ML: -14943.712589051003

Maximum Likelihood Estimation: -14943.712589051003

Log-Likelihood -14943.712589051003

Class 2

Class 2 mu: [[-0.00470504] [-0.00937712] [0.6559566] [-0.65899175] [-0.04389534] [0.02496492]]

Class 2 C: [[1.4480951e+00 -1.4722235e-02 5.5701016e-03 1.5741589e-02 1.9497097e-02 -1.7668604e-04] [-1.4722235e-02 5.5339074e-01 -1.1216865e-02 -9.0647331e-03 -1.4658980e-02 1.6349208e-02] [5.5701016e-03 -1.1216865e-02 5.5748022e-01 2.7560966e-02

-3.7696741e-03 -1.4597691e-02] [1.5741589e-02 -9.0647331e-03 2.7560966e-02 5.6965697e-01 -1.1698328e-02 3.4993179e-02] [1.9497097e-02 -1.4658980e-02 -3.7696741e-03 -1.1698328e-02 1.3420180e+00 1.6945411e-02] [-1.7668604e-04 1.6349208e-02 -1.4597691e-02 3.4993179e-02 1.6945411e-02 1.3037186e+00]]

Difference between mu: 0.0

Difference between C: 0.0

Maximum Likelihood Estimation - ML: -16227.483891232197

Maximum Likelihood Estimation: -16227.483891232197

Log-Likelihood -16227.483891232197

Validation data Class 1

Class 1 mu: [[0.02181397] [0.02399015] [-0.6947283] [0.6513403] [0.0527217] [0.00920085]]

Class 1 C: [[5.0585079e-01 1.3110590e-02 -1.2581978e-02 5.3225174e-03 1.6512774e-02 2.2454146e-02] [1.3110590e-02 1.3677320e+00 1.8263675e-02 -4.8120529e-03 4.5328110e-02 -6.2971443e-02] [-1.2581978e-02 1.8263675e-02 5.1873195e-01 4.3107741e-02 -2.0300722e-02 -2.4406696e-04] [5.3225174e-03 -4.8120529e-03 4.3107741e-02 5.2425003e-01 1.1466244e-02 2.1252098e-02] [1.6512774e-02 4.5328110e-02 -2.0300722e-02 1.1466244e-02 6.4694214e-01 3.7016172e-03] [2.2454146e-02 -6.2971443e-02 -2.4406696e-04 2.1252098e-02 3.7016172e-03 7.4199921e-01]]

Difference between mu: 0.0

Difference between C: 0.0

Maximum Likelihood Estimation - ML: -7243.107427638617

Maximum Likelihood Estimation: -7243.107427638617

Log-Likelihood -7243.107427638617

Class 2

Class 2 mu: [[0.0109708] [-0.00683072] [0.68367106] [-0.67452973] [-0.03741501] [0.0219]]

Class 2 C: [[1.394594 0.04672578 -0.02116859 0.00787587 -0.0217899 -0.07246453] [0.04672578 0.62770224 -0.00339103 -0.00234733 0.01002544 -0.01969187] [-0.02116859 -0.00339103 0.5313556 -0.0208077 0.02126161 -0.02575587] [0.00787587 -0.00234733 -0.0208077 0.5207802 0.01469356 -0.00856059] [-0.0217899 0.01002544 0.02126161 0.01469356 1.2695773 -0.06093844] [-0.07246453 -0.01969187 -0.02575587 -0.00856059 -0.06093844 1.2538664]]

Difference between mu: 0.0

Difference between C: 0.0

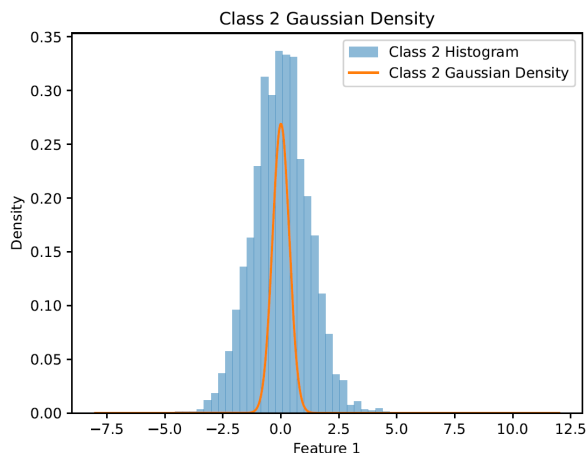
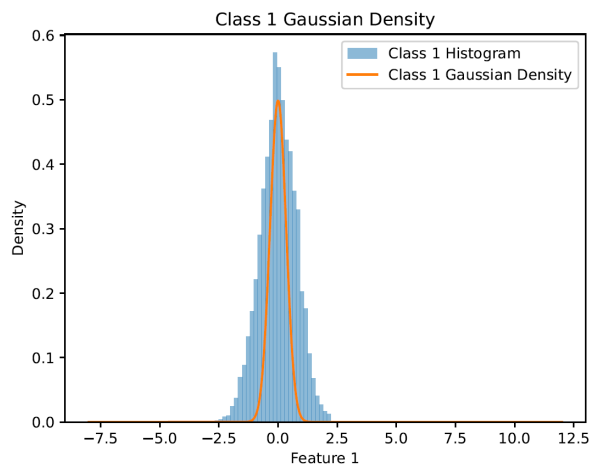
Maximum Likelihood Estimation - ML: -8094.842732690457

Maximum Likelihood Estimation: -8094.842732690457

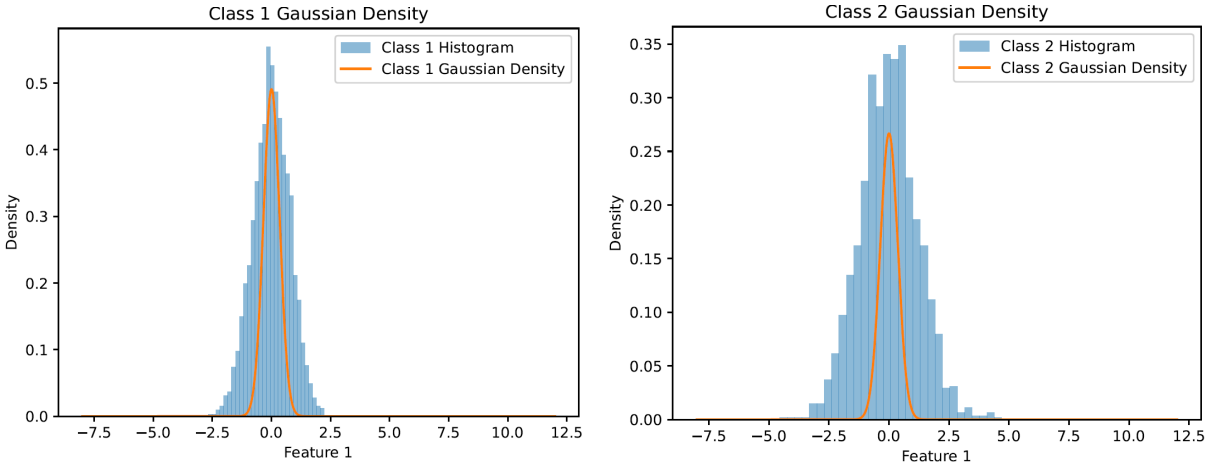
Log-Likelihood -8094.842732690457

Gaussian density plots

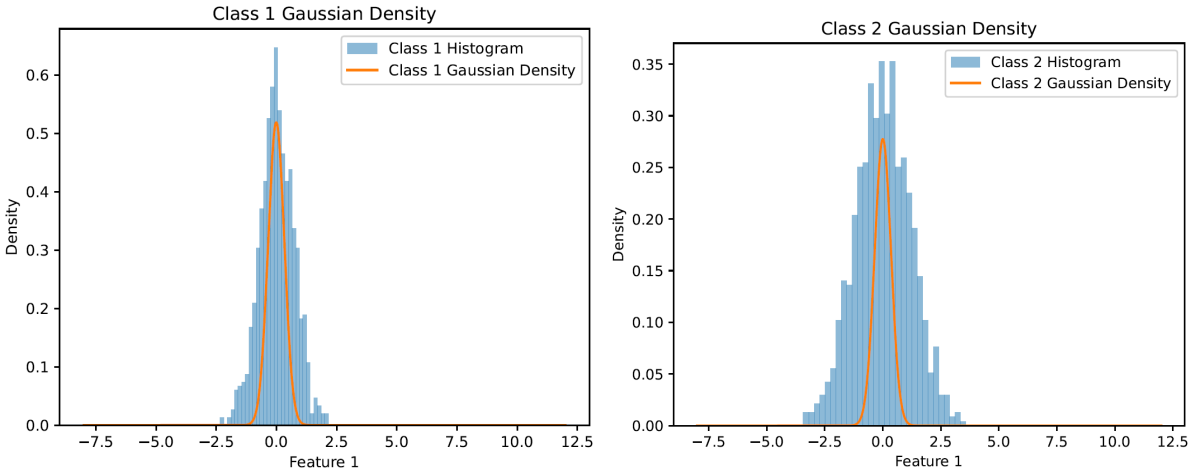
All data



Training data

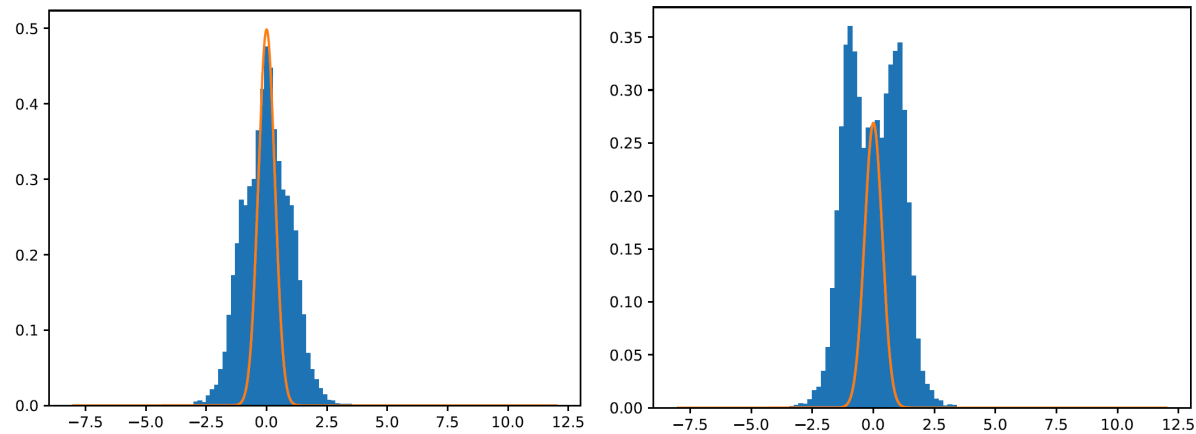


Validation data

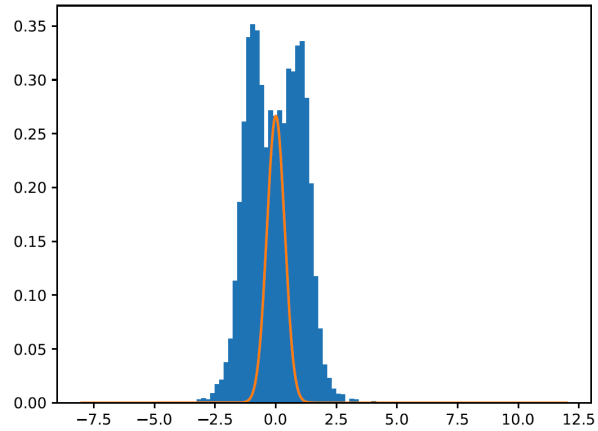
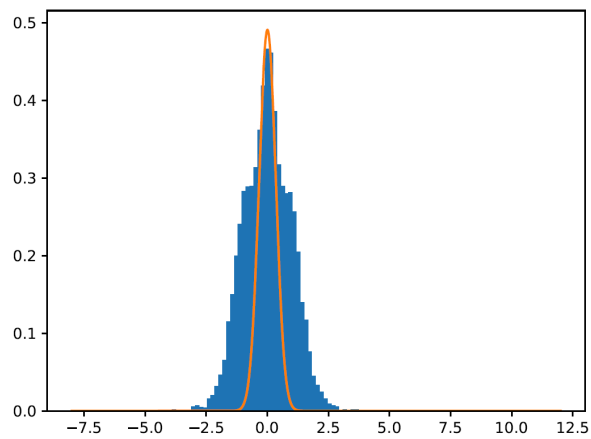


Likelihood plots

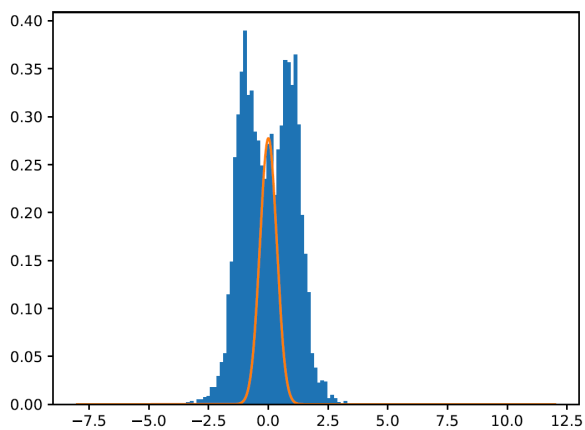
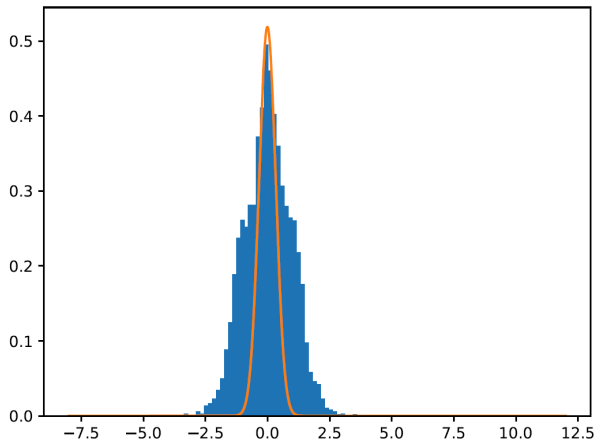
All data



Training data



Validation data



Gaussian Distribution Analysis

Statistical Properties

Overall Dataset

- Mean: Computed for all features across the entire dataset.
- Variance: Estimated for each feature to observe dispersion.

Class-wise Analysis

- Class 1 (Fake)
 - Mean: Computed separately for this class.
 - Variance: Evaluated to check intra-class dispersion.
- Class 2 (Genuine)
 - Mean: Analyzed independently.
 - Variance: Compared to Class 1 for insights.

Training vs Validation Comparison

To analyze model performance, we compare statistics between training and validation datasets:

Training Data

- Feature distributions are visualized to ensure Gaussian properties.
- Mean and variance trends are checked for consistency.

Validation Data

- Similar statistical measures are computed.
- Differences with training data are evaluated.

Comparisons

Full Dataset vs Individual Classes

- The overall mean and variance of the full dataset are compared against individual class statistics.
- Observations on whether class distributions significantly deviate from the overall distribution.

Training vs Validation

- Differences in feature means and variances between training and validation data.
- Checking for data shifts or potential overfitting in the training phase.

Conclusion

- The Gaussian assumption is validated by assessing the distributions.
- Discrepancies in feature distributions suggest potential preprocessing adjustments.
- Training and validation data distributions provide insights into generalization and possible biases.